

Name: _____

FRACTIONS, DECIMALS AND PERCENTAGES, (PERCENTAGE CHANGE)

Date: _____

LO: I can calculate percentage increase, decrease and percentage change.

What is percentage change?

Percentage change measures how much a value has **increased or decreased** as a percentage of the original.

Formula: % change = (change ÷ original) × 100. Always divide by the ORIGINAL amount, not the new amount.

Examples

●	£20	£25 = +25%	increase
●	£40	£30 = -25%	decrease
●	10	12 = +20%	(2÷10)×100
●	£20	£30 = +33%	should be +50%
●	Divide by NEW value	always divide by ORIGINAL	

Key idea

% change = (new - old) ÷ old × 100. Positive = increase, negative = decrease.

$$\frac{\text{change}}{\text{original}} \times 100 = \text{\% change}$$

Always divide by ORIGINAL

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - increase

Price rises from £40 to £50. Find the % increase.

$$\begin{aligned}\text{Change} &= 50 - 40 = 10 \\ 10 \div 40 &= 0.25 \\ 0.25 \times 100 &= 25\%\end{aligned}$$

= 25% increase

Always divide the change by the ORIGINAL price (£40).

Example 2 - decrease

Population drops from 800 to 600. Find the % decrease.

$$\begin{aligned}\text{Change} &= 800 - 600 = 200 \\ 200 \div 800 &= 0.25 \\ 0.25 \times 100 &= 25\%\end{aligned}$$

= 25% decrease

The change is always positive; the 'decrease' describes the direction.

Example 3 - from amounts

A coat costs £80 in a sale. The original price was £100. % change?

$$\begin{aligned}\text{Change} &= 80 - 100 = -20 \\ -20 \div 100 &= -0.20 \\ -0.20 \times 100 &= -20\%\end{aligned}$$

= 20% decrease

Negative answer means a decrease.

Example 4 - growth

Plant grows from 12 cm to 18 cm. % growth?

$$\begin{aligned}\text{Change} &= 18 - 12 = 6 \\ 6 \div 12 &= 0.5 \\ 0.5 \times 100 &= 50\%\end{aligned}$$

= 50% growth

Divide the increase by the original (12 cm).

YOUR TURN – PRACTICE (deliberate practice)

1. Find the change

Calculate the change (new - old).

- a) £50 £65
- b) £90 £75
- c) 200 240
- d) 150 120

2. % increase

Calculate the percentage increase. Use $(\text{change} \div \text{old}) \times 100$.

- a) £20 £25
- b) £40 £50
- c) 60 90
- d) 200 240

3. % decrease

Calculate the percentage decrease.

- a) £40 £30
- b) £100 £85
- c) 150 120
- d) 80 60

4. Word problems

State whether each is an increase or decrease, and find the %.

- a) Salary £25k £30k
- b) Houses 400 480
- c) Bill £80 £64
- d) Speed 50 35 mph

5. Mixed % change

Find the percentage change. State increase or decrease.

- a) 12 15
- b) £90 £108
- c) 60 45
- d) £500 £575
- e) 120 90

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) £20 £25 is a 25% increase ☐

Correct answer: _____

Reason: _____

b) Divide by the new value, not the old ☐

Correct answer: _____

Reason: _____

c) £100 £80 is a 20% decrease ☐

Correct answer: _____

Reason: _____

d) 200 240 is a 40% increase ☐

Correct answer: _____

Reason: _____

e) % change can be negative for a decrease ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A shop sells a TV for **£500**. In a sale they reduce it to **£425**. After the sale they raise it back to **£480**. Find the % decrease in the sale and the % increase after the sale. Are they the same number? Explain why or why not.

Expression: _____

Simplified: _____

b) A car worth **£12 000** loses **15% of its value** in year 1, then another **10%** in year 2. What is the car worth after 2 years? What's the total % change?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

